

EN 842300 · INTERACTIVE WEB PROGRAMMING · KHON KAEN
UNIVERSITY

DME Explorer

Project report — motivation, scope, methodology, and
interface prototype for an interactive web guide to the
DME program at Khon Kaen University.

Project: DME Explorer — Digital Media Engineering Program Guide

Live site: dmeexplorer.vercel.app

Repository: github.com/MONNNNNNNNNNN/project_webinteract

Final presentation / demo day: 21 October 2026

Motivation

Prospective and first-year Digital Media Engineering (DME) students at Khon Kaen University's Faculty of Engineering currently have to piece their understanding of the program together from several disconnected sources: static faculty web pages, PDF curriculum documents (course maps, TQF.2 specifications), and word of mouth from seniors. There is no single place that shows the 4-year study plan, the cost of attendance by student type, what kind of jobs the degree leads to, what past students have actually built, and what the program's facilities look like — all at once, kept current, and easy to explore.

This gap is the direct motivation for DME Explorer: an interactive web guide built for EN 842300 (Interactive Web Programming) that consolidates this information into one resource, grounded in real data pulled from KKU's own official sources rather than illustrative placeholder content.

Related Apps

No single existing tool covers everything DME Explorer targets, but several categories of existing products cover parts of the same problem space:

Product	Category	What it covers	Gap vs. DME Explorer
University of Illinois Course Explorer	Course catalog search	Searchable course descriptions, prerequisites, and scheduling across an entire university	Generic and broad — not curated for one program's specific track structure, no tuition/career/3D content
MyU Student Portal (mobile)	Enrolled-student portal	Class schedules, fees, exam timetables/results for currently enrolled students	Built for students already enrolled, not prospective ones deciding whether to apply; no public marketing/overview content
"University Explorer" study-abroad app	Cross-university discovery	Search universities/programs across 13+ countries by field, degree type, tuition	Broad and shallow by design — a directory, not a deep single-program resource with live data or interactivity
JobsDB / LinkedIn Jobs (Thailand)	General job search	Real, live job postings across all industries and companies in Thailand	Not filtered to DME-relevant career tracks (3D & Animation, Game Dev, AI & Data, Software) or tied to the curriculum that produces those skills
en.kku.ac.th/web/en/beng-dme (official KKU program page)	Faculty program page	The authoritative source for program description, credit structure, and tuition — the actual baseline this project builds on	Static text/PDF only — no interactive calculator, no course-by-course browsing, no live job data, no chatbot, no 3D preview

DME Explorer's differentiation is combining DME-specific curated content with live external data (job listings via JSearch, an AI FAQ chatbot via Claude), interactive tools (tuition calculator, expandable course catalog), and a first-person 3D walkthrough of the CDLC facility — in one place, aimed specifically at prospective and first-year DME applicants.

Objectives

1. Consolidate DME program information — curriculum, tuition, careers, and student work — into a single interactive resource for prospective and first-year students.
2. Present the full 4-year, 120-credit study plan and the four major elective tracks (AI, Digital Media, Interactive, Software) in a way that's easy to browse rather than read off a static PDF.
3. Give a transparent, real-time cost breakdown by student type (Thai / Mekong Region / International), sourced from the official program page.
4. Surface job-market data relevant to DME graduates, filterable by career interest area.
5. Showcase real DME student work and achievements, not fabricated examples.
6. Provide an AI chatbot that can answer common admissions/program questions instantly, scoped to DME facts only.
7. Offer a first-person 3D preview of the CDLC facility so prospective students get a sense of the physical space before visiting.
8. Build all of the above on a modern, accessible, responsive interface (light/dark mode, mobile-first, smooth motion) rather than a static brochure site.

Scope

In scope

- **Curriculum Roadmap** — full 4-year study plan, per-year credit breakdown, all-courses catalog grouped by the official 6 credit categories, and the 4 major elective tracks, each course expandable to show real official descriptions and prerequisites.
- **Tuition & Fees** — interactive calculator across student type and billing period (per-semester / full 4 years), with a full cost breakdown (tuition, health insurance, accommodation, living costs).
- **Career Explorer** — job listings filterable by interest track, live via the JSearch API when available, with a simulated fallback pointing to real company career pages.
- **Student Projects & Competition** — a showcase of real DME student work and awards, clearly distinguished from placeholder entries pending more submissions.
- **3D World** — a first-person walkthrough of a CDLC facility model, built directly in-app.
- **About** — program overview plus a full directory of real teaching staff (bios, photos, specialties) sourced from the official department directory.
- **Contact** — real faculty contact channels (International Affairs Division, general faculty line, address, social).
- **DME Assistant** — a floating chatbot scoped strictly to DME/KKU topics.
- **Admin panel** — a stubbed login/dashboard for managing chatbot FAQ content (simulated auth for now).

Out of scope

- Alumni stories / alumni admin panel — considered in an earlier draft of the project, cut from the final scope.
- Full production authentication (Supabase Auth wiring is planned but not yet implemented — admin login currently uses a simulated session).
- Multilingual (Thai-language) UI — noted as future work.

Platform constraints

Hosted on Vercel's Hobby tier, which caps serverless function execution at 10 seconds — this is why the chatbot uses Claude Haiku (fast, cheap) rather than a slower reasoning model. The planned Supabase database is also on a free tier that pauses after 7 days of inactivity, which the team accounts for ahead of any live demo.

Tools & Technology

Layer	Technology	Why
Frontend framework	React 18 + Vite 5	Fast dev server, component-based UI, wide ecosystem
Styling	Tailwind CSS 3	Consistent design tokens, rapid iteration, built-in dark-mode variant support
Routing	React Router 6	Client-side navigation across 9 routes
Animation	Framer Motion	Page transitions, accordions, carousels, active-nav indicator
3D rendering	Three.js + React Three Fiber + drei	First-person walkthrough of the CDLC model (.glb), pointer-lock + WASD controls on desktop, orbit controls on touch
Icons	lucide-react	Consistent, lightweight SVG icon set
Backend	Vercel Serverless Functions (Node.js)	No server to manage, scales with the Vercel Hobby plan, colocated with the frontend deploy
Planned database	Supabase (Postgres)	Auth + FAQ storage for the admin panel (not yet wired in)
External APIs	JSearch (RapidAPI), Anthropic Claude API	Live job-market data; AI chatbot
Hosting / CI	Vercel (Hobby tier)	Git-connected deploys, zero-config for a Vite app
Version control	Git + GitHub	Full commit history, collaborative editing

API Implemented

Endpoint	Method	Purpose	Fallback behavior
/api/careers	GET	Job listings filterable by ?interest= (3D & Animation / Game Dev / AI & Data / Software), sourced from JSearch (RapidAPI)	Falls back to a 16-entry simulated pool linking to each company's real careers page; results are shuffled server-side per request (Cache-Control: no-store) so a refresh looks meaningfully different
/api/chat	POST	DME Assistant chatbot, backed by Anthropic's Claude Haiku 4.5 with a system prompt grounded in real program facts (credits, tuition, contacts) and an explicit instruction to decline off-topic questions	Falls back to a small hardcoded FAQ matcher covering the same core facts — zero marginal cost, still DME-scoped
/api/admin/login	POST	Admin session creation	Simulated demo credential; real Supabase Auth wiring is planned, not yet built
/api/admin/session	GET	Session check for the admin dashboard	—
/api/admin/logout	POST	Clears the admin session	—
/api/admin/faqs	GET / POST / DELETE	CRUD for chatbot FAQ entries from the admin dashboard	In-memory store — resets on server restart until Supabase is wired in

Design principle applied across every integration: an unavailable or rate-limited external API must never break the page — the UI always renders something useful, clearly labeled as simulated when it is.

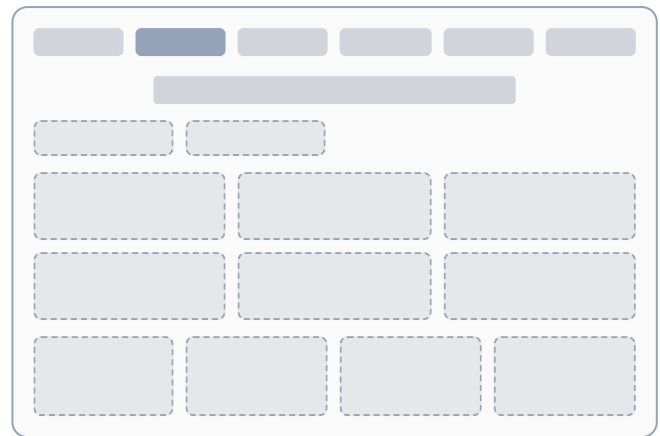
Interface Prototype

Low-fidelity (sketch)

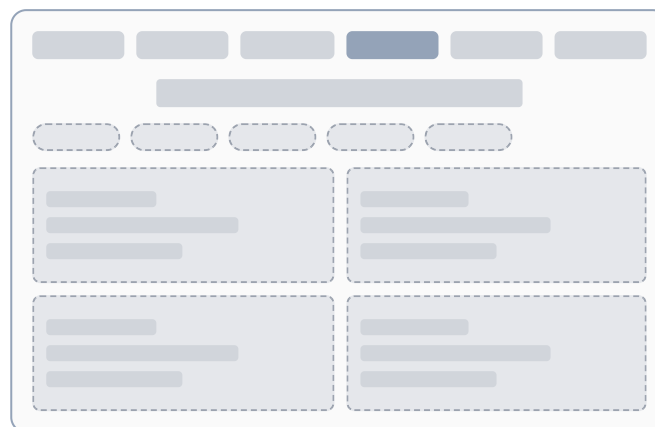
Early layout sketches for the three most interaction-heavy pages, before any visual styling — used to agree on structure (navigation, content blocks, card grids) before writing real UI code.



Home — hero + auto-rotating news carousel



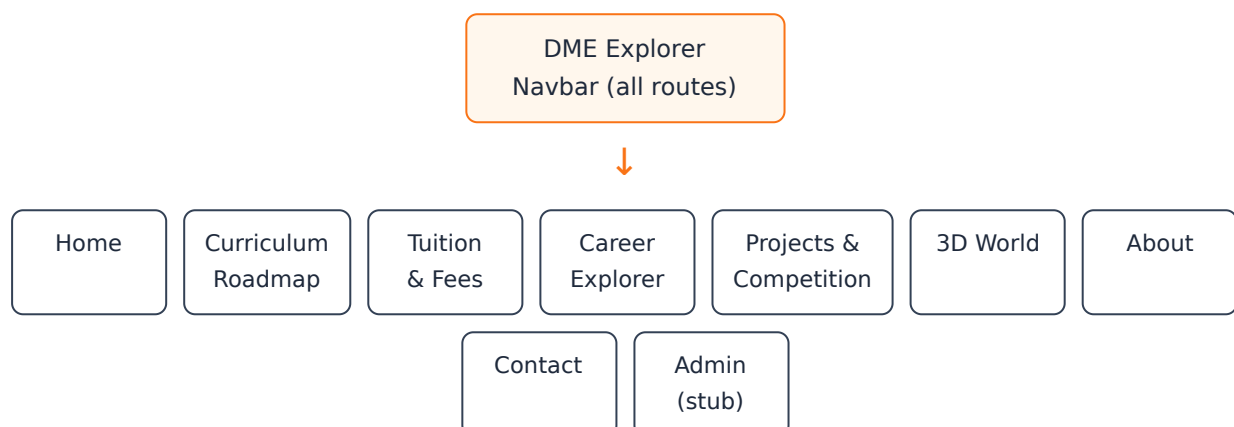
Curriculum — tab switcher + credit breakdown cards + year cards



Career Explorer — filter chips + job listing cards

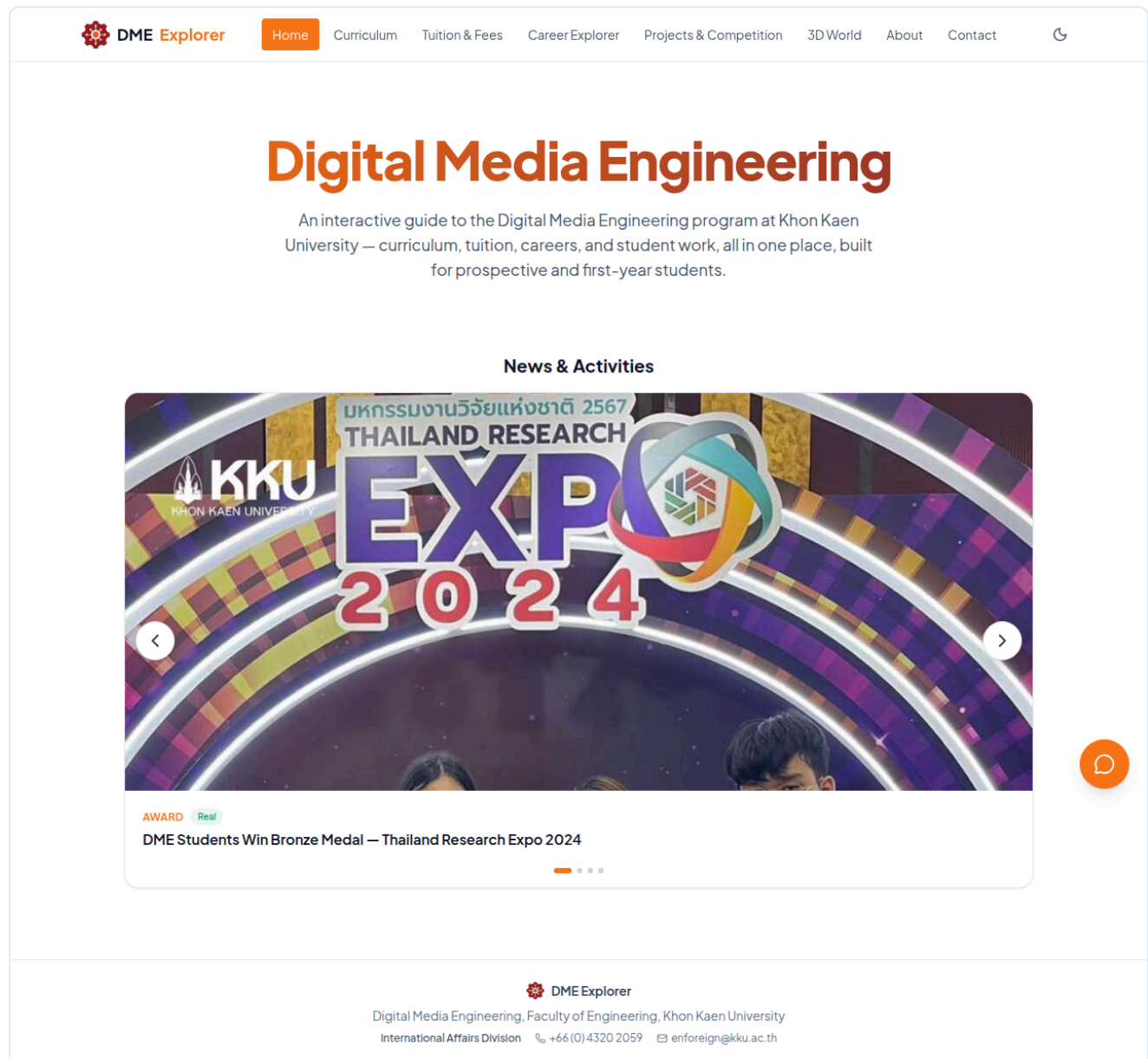
Web Overview Diagram

Site map: every route hangs off a single persistent navbar, with no dead ends — every page is reachable in one click from anywhere else in the app.



High-fidelity

The interface has been fully implemented, so the high-fidelity representation below is captured directly from the live, deployed application (dmeexplorer.vercel.app) rather than a separate static mockup — it is the actual working product, in both light mode (default) and responsive mobile layout.



Home — hero, auto-rotating real news/activity carousel (drag or arrows to navigate)

Digital Media Engineering

An interactive guide to the Digital Media Engineering program at Khon Kaen University — curriculum, tuition, careers, and student work, all in one place, built for prospective and first-year students.

News & Activities



THAILAND RESEARCH
EXPO
2024

นักศึกษาวิศวกรรมสื่อดิจิทัล มข.
คว้ารางวัลเหรียญทองแดง

AWARD Real

DME Students Win Bronze Medal —
Thailand Research Expo 2024

Curriculum Roadmap

120 credits over 4 years (Cooperative Education track).

[Curriculum](#)[Course](#)

What you'll study

Digital Media Engineering students specialize in developing, implementing, and optimizing technology systems for creating, processing, delivering, and displaying digital content.

Core responsibilities: designing applications, implementing streaming technologies, developing interactive experiences, and building asset management systems.

Key competencies: programming for audio/video, streaming protocols, interactive media development, user experience design, and virtual/augmented reality technologies.

Credit breakdown

30**General Education**

Language 12 · Humanities/Social Sciences 6 ·
Math/Sciences 12

15**Basic Engineering**

Fundamental courses

36**Core Engineering**

Compulsory major courses

27**Elective Engineering**

Min. — AI / Digital Media / Interactive / Software tracks

6**Field Experience**

Practical training / co-op

6**Free Elective**

Min. — any faculty

Year by year

Year 1**32**

credits this year · 12 courses

Year 2**32**

credits this year · 12 courses

Year 3**32**

credits this year · 12 courses

Year 4**24**

credits this year · 3 courses





Tuition & Fees

Select your student type to see a transparent breakdown of academic and living costs, sourced from the official program page.

SEMESTER FEE
Thai Students
฿45,000

SEMESTER FEE
Mekong Region
฿50,000

SEMESTER FEE
International
฿65,000

Fee Breakdown

Detailed costs for Thai Students.

Per Semester

Full 4 Years

ACADEMIC COSTS

Tuition Fee **MANDATORY** **฿45,000**

Summer Training **ONE-TIME** **฿2,500**

GRAND TOTAL (PER SEMESTER)

Includes mandatory academic fees + 4 months of estimated living costs.

฿45,000



Tuition & Fees — real-time calculator by student type and billing period

Career Explorer

Live, currently-open job listings for DME graduates, filterable by interest area. Showing simulated data — set JSEARCH_API_KEY for live listings.

All

3D & Animation

Game Dev

AI & Data

Software

Refresh

AI & DATA

Machine Learning Engineer, Intern

AIS

Bangkok, Thailand

3D & ANIMATION

3D Environment Artist

Yggdrasil Group

Bangkok, Thailand

SOFTWARE

Backend Developer, New Grad

LINE Thailand

Bangkok, Thailand

SOFTWARE

Frontend Developer (React)

Bitkub

Bangkok, Thailand

GAME DEV

Game Designer

Corecell Technology

Bangkok, Thailand

AI & DATA

Data Analyst

SCB 10X

Bangkok, Thailand (Hybrid)

SOFTWARE

Software Engineer

Agoda

Bangkok, Thailand

GAME DEV

Technical Artist

Yggdrasil Group

Bangkok, Thailand

SOFTWARE

Full-Stack Developer

True Digital Group

Bangkok, Thailand

AI & DATA

Data Engineer

Sertis

Bangkok, Thailand

3D & ANIMATION

3D Character Artist

Riot Games

Bangkok, Thailand (Hybrid)

3D & ANIMATION

Junior Animator

Kantana Animation Studio

Bangkok, Thailand

AI & DATA

AI Research Engineer

Amity Solutions

Bangkok, Thailand (Hybrid)

3D & ANIMATION

VFX Compositor

The Monk Studios

Bangkok, Thailand

GAME DEV

Unity Developer (Junior)

Extend Interactive

Remote (Thailand)

GAME DEV

Gameplay Programmer

Garena

Bangkok, Thailand

Student Projects & Competition

A sample of work, activities, and competition results from DME students. Entries marked "Real" are sourced from official KKU coverage; the rest are placeholder entries pending real project media and links.



Award Real

Thailand Research Expo 2024 — Bronze Medal

DME students won a Bronze Medal at the Higher Education Innovation Stage, Thailand Research Expo 2024 (===== 2567).



Field Study Real

Studio Visit — The Monk Studios, Igdrasil Group, Zurreal Studio

March 7–9, 2024: 36 third-year DME students visited three digital media companies to build real-world industry experience alongside classroom learning.



Digital Media

Interactive Album Story

A 3D animated short combining character rigging and real-time rendering, produced as a Digital Media Studio capstone.



Interactive

Campus Quest

A game-dev orientation project that turns the KKU campus into an explorable 2D game for incoming freshmen.



Software / AI

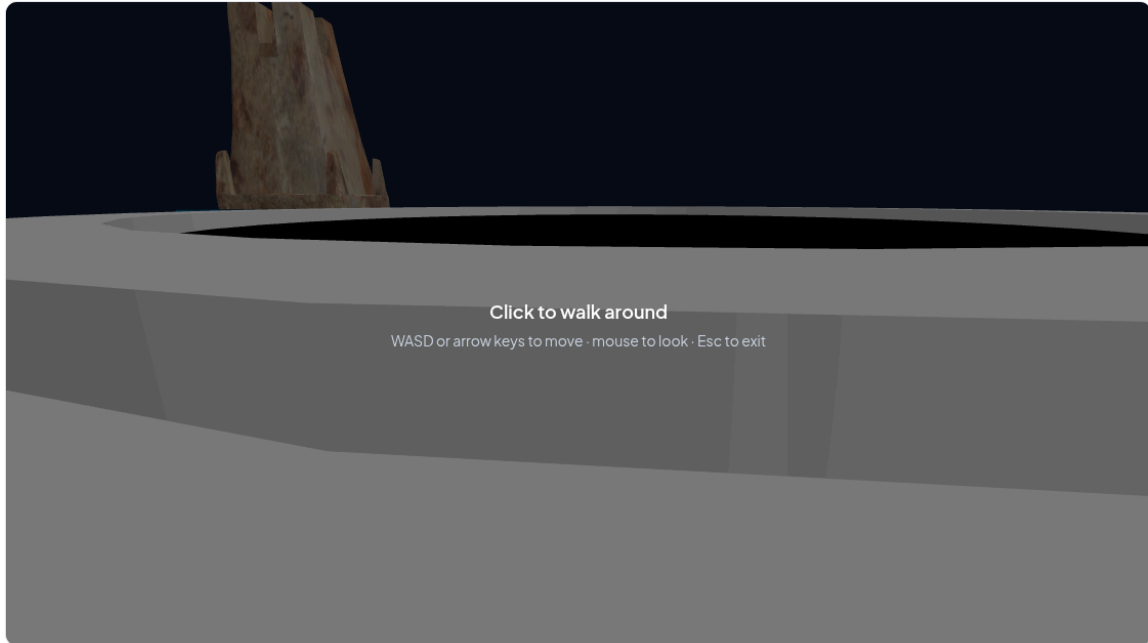
DME FAQ Assistant

An early prototype chatbot answering common DME admissions questions, built with a lightweight NLP pipeline.



3D World — CDLC Simulation

Test only. Real coming soon Touch devices: drag to look around.



3D World — first-person CDLC walkthrough (React Three Fiber, WASD + pointer-lock on desktop)



DIGITAL MEDIA
ENGINEERING

About DME

Digital Media Engineering at Khon Kaen University's Faculty of Engineering.

What is DME?

Digital Media Engineering (DME) is an international undergraduate program at Khon Kaen University's Faculty of Engineering that blends software engineering fundamentals with digital media production — 3D/animation, interactive media, AI, and game/software development.

What do we learn?

Vision & Mission

Why choose DME KKU?

Lecturer

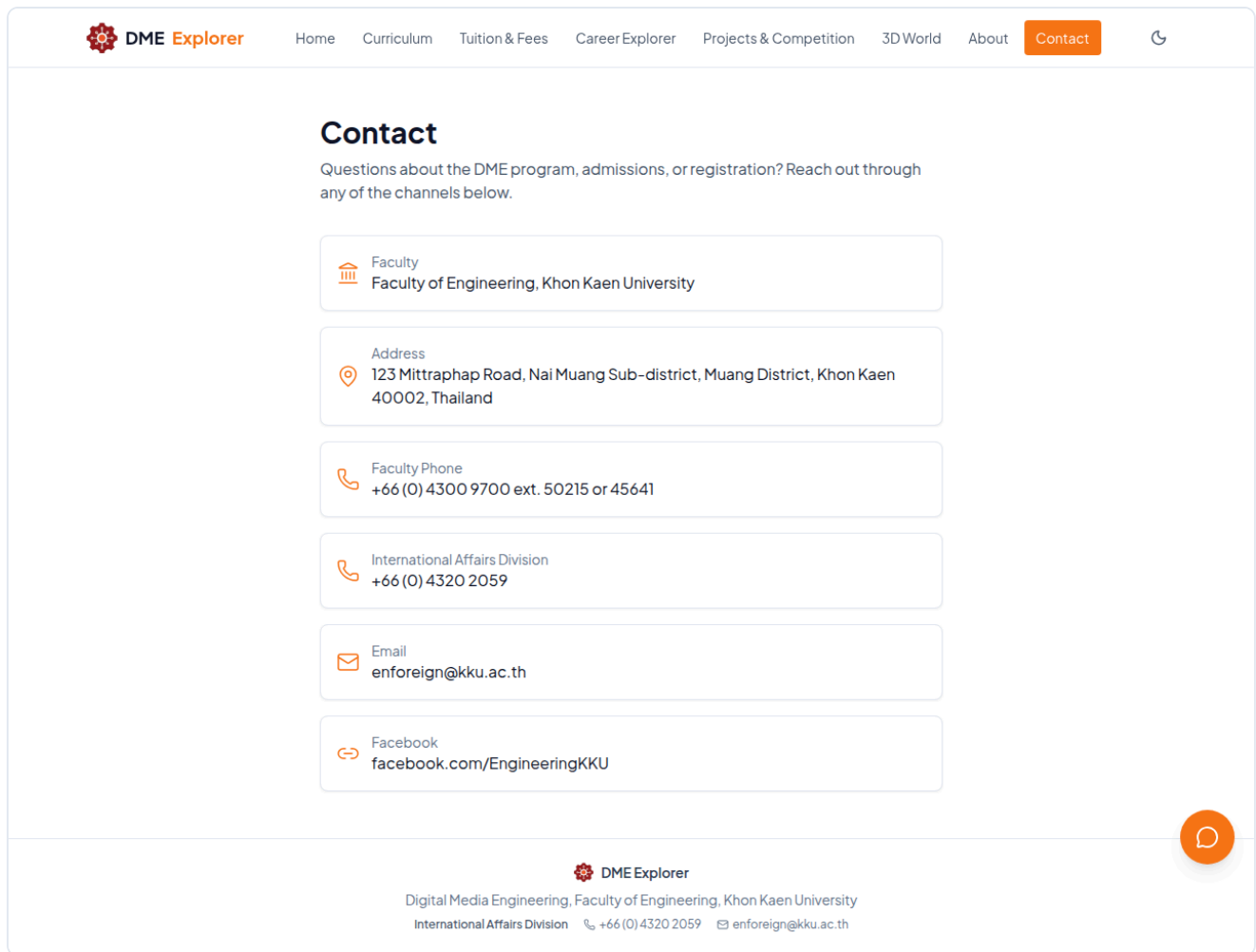


DME Explorer

Digital Media Engineering, Faculty of Engineering, Khon Kaen University
International Affairs Division ☎ +66 (0) 4320 2059 ✉ enforeign@kku.ac.th



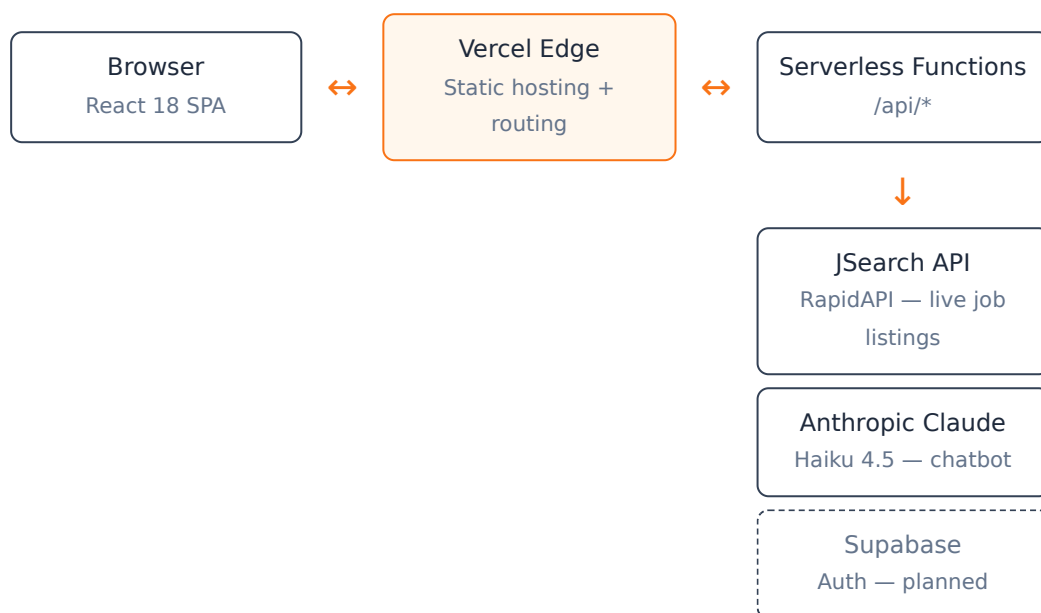
About — accordion facts plus a real teaching-staff directory



Contact — real faculty contact channels

System Flow

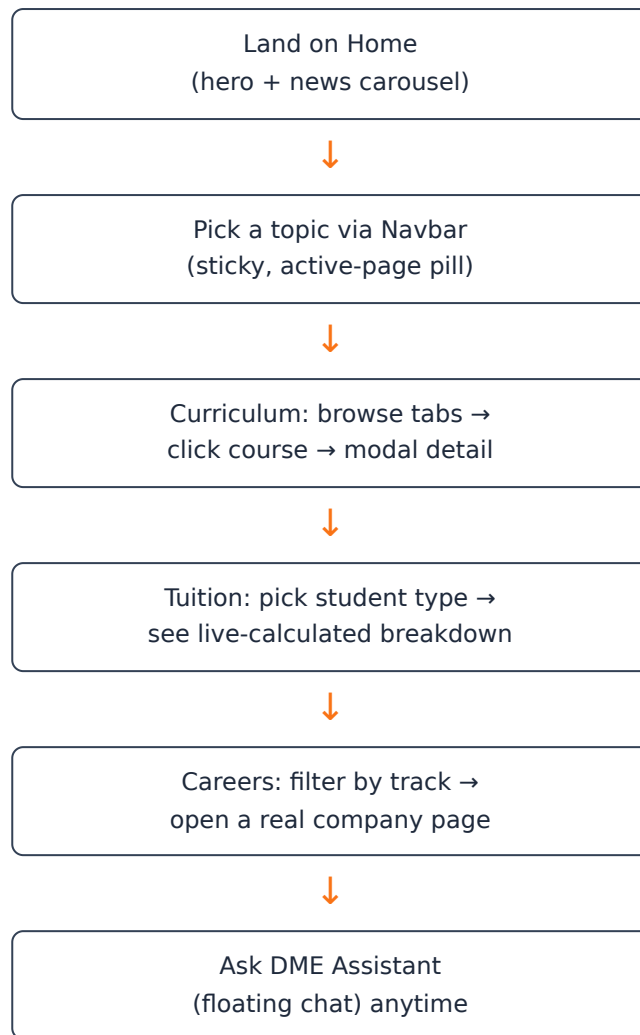
How a request moves through the stack, and what happens when an external dependency is unavailable.



Every external call has a graceful simulated fallback (mock data, DME-scoped hardcoded FAQ) when a key/quota isn't available — the app never shows a broken state.

Web App Flow

The typical path a prospective student takes through the app on a first visit.



Methodology

The project followed an iterative, feature-by-feature build rather than a single big-bang release:

1. **Source-first data gathering.** Before building any content page, the underlying real data was sourced first — the official KKU program page, the department staff directory, the official TQF.2 curriculum specification document, the university registrar, and a prior semester's capstone report used as a secondary reference for tuition figures. Placeholder/demo data was treated as temporary scaffolding, not a final state.
2. **Page-by-page implementation.** Static, self-contained pages (Home, Curriculum, Tuition, About, Contact) were built first with real content and no backend dependency, followed by pages requiring live external data (Career Explorer, DME Assistant) built with a simulated fallback from day one so the app never depends on an external key being present to function.
3. **Continuous deployment.** Every change was built and verified locally, committed to Git with a descriptive message, pushed to GitHub, and deployed to Vercel production — keeping the live site and the repository in sync throughout development rather than batching changes into infrequent releases.
4. **Feedback-driven iteration.** After each deployment, the live site was reviewed (screenshots, direct interaction, and screen recordings of real usage) and issues were reported back with specific reproduction steps. Several real production bugs were found this way and root-caused rather than patched superficially — for example, a mobile Safari overscroll bug, a missing scroll-reset on route change, and a framer-motion shared-layout animation glitch that only appeared when scrolling and navigating together.
5. **Data-accuracy discipline.** Where real official data was found to differ from earlier estimates (for example, tuition figures and per-year credit totals), the site was corrected and the source re-verified rather than leaving the discrepancy unresolved.

Timeline

Phase	Period	Milestones
1 — Foundation	Mid July 2026	Project scaffold (Vite + React + Tailwind), real curriculum & tuition data sourced, stub pages for all routes
2 — Backend & identity	Mid-late July 2026	Simulated backend for chat/careers/admin, DME branding and logo, light/dark theme, page-transition polish
3 — Real data integration	Late July 2026	Live KKU sourcing for curriculum, tuition, staff, contacts, and news; accordion/lightbox/carousel UX; multiple production bug fixes
4 — Depth features	Early August 2026	Official course-by-course descriptions extracted from the university's TQF.2 curriculum document; first-person 3D CDLC walkthrough built with React Three Fiber
5 — Hardening & docs	August - September 2026	Supabase admin authentication, resolving live job-API quota constraints, cross-device QA, documentation
6 — Final presentation	21 October 2026	Demo day — live walkthrough of the deployed site

Conclusions

DME Explorer consolidates what used to be scattered across static faculty pages, PDF curriculum documents, and word of mouth into one interactive, continuously-updated resource for prospective and first-year Digital Media Engineering students at Khon Kaen University. Its main differentiators against generic university portals and directory apps are: DME-specific curated content sourced from official university documents, live job-market integration, an AI-scoped chatbot, an interactive tuition calculator, a fully browsable course catalog with real official descriptions, and a first-person 3D walkthrough of the CDLC facility — all combined in a single, modern, responsive interface.

Development followed a source-data-first, continuously-deployed, feedback-driven process, which surfaced and fixed a number of real usability and data-accuracy issues before they would have reached an actual prospective student.

Remaining work before the final presentation includes wiring real Supabase authentication for the admin panel, resolving the JSearch API's request quota so live job data is available for the demo (with the simulated fallback as a documented contingency), and — as future work beyond this course — Thai-language support and extending the same approach to other KKU programs.